

NET High-Yield FSc Mathematics Formula and Trick Book

Math-only formula sheet covering the topic families appearing in the attached MCQ papers

Designed for rapid revision, MCQ elimination, and 2-day crash preparation

1. Number System and Algebra Basics

Indices and Surds

$$\begin{aligned}a^m a^n &= a^{m+n}, & \frac{a^m}{a^n} &= a^{m-n}, & (a^m)^n &= a^{mn} \\(ab)^n &= a^n b^n, & \left(\frac{a}{b}\right)^n &= \frac{a^n}{b^n}, & a^0 &= 1, & a^{-n} &= \frac{1}{a^n} \\a^{1/n} &= \sqrt[n]{a}, & a^{m/n} &= \sqrt[n]{a^m}, & \sqrt{ab} &= \sqrt{a}\sqrt{b} \\(a+b\sqrt{c})(a-b\sqrt{c}) &= a^2 - b^2c, & \frac{1}{a+\sqrt{b}} &= \frac{a-\sqrt{b}}{a^2-b}\end{aligned}$$

Factorization and Absolute Value

$$\begin{aligned}a^2 - b^2 &= (a-b)(a+b), & a^3 \pm b^3 &= (a \pm b)(a^2 \mp ab + b^2) \\(a \pm b)^2 &= a^2 \pm 2ab + b^2 \\|x-a| < b &\Rightarrow a-b < x < a+b, & |x-a| > b &\Rightarrow x < a-b \text{ or } x > a+b\end{aligned}$$

Tricks: Use factorization before expansion. For $|x-a| < b$, write interval directly. Rationalize before matching options.

2. Logarithms and Exponents

$$\begin{aligned}\log_a xy &= \log_a x + \log_a y, & \log_a \frac{x}{y} &= \log_a x - \log_a y, & \log_a x^n &= n \log_a x \\ \log_a a &= 1, & \log_a 1 &= 0, & a^{\log_a x} &= x, & \log_a x &= \frac{\log_b x}{\log_b a} \\ a^x &= a^y \Rightarrow x = y, & a^x &= b \Rightarrow x = \log_a b\end{aligned}$$

Tricks: Check log domain: argument > 0 , base > 0 , base $\neq 1$. Convert to the same base.

3. Sets, Relations, Functions

$$\begin{aligned}A - B &= A \cap B', & (A \cup B)' &= A' \cap B', & (A \cap B)' &= A' \cup B' \\ n(A \cup B) &= n(A) + n(B) - n(A \cap B), & n(A - B) &= n(A) - n(A \cap B) \\ n(A \cup B \cup C) &= n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C) \\ R \subseteq A \times B, & (a, a) \in R \text{ reflexive, } & (a, b) \in R &\Rightarrow (b, a) \in R \text{ symmetric} \\ & (a, b), (b, c) \in R &\Rightarrow (a, c) \in R \text{ transitive} \\ (g \circ f)(x) &= g(f(x)), & (f \circ g)(x) &= f(g(x)), & g \circ f &\neq f \circ g \text{ usually} \\ y = f(x) &\Rightarrow \text{solve for } x \Rightarrow f^{-1}(x) \\ \sqrt{x-a} : x &\geq a, & \frac{1}{x-a} : x &\neq a, & \log(x-a) : x &> a\end{aligned}$$

Tricks: In $A - B$, remove from A only. For $g \circ f$, apply f first. Domain traps: root, denominator, log.

4. Groups and Binary Operations

$$\begin{aligned}* : G \times G &\rightarrow G, & a, b \in G &\Rightarrow a * b \in G \text{ closure} \\ (a * b) * c &= a * (b * c), & a * e &= e * a = a, & a * a^{-1} &= a^{-1} * a = e \\ a * b &= b * a \Rightarrow \text{abelian}\end{aligned}$$

Tricks: Identity depends on the operation. Inverse is not always $1/a$. Associativity is grouping; commutativity is order.

5. Complex Numbers

$$\begin{aligned}z &= a + bi, \quad \Re(z) = a, \quad \Im(z) = b \\i^1 &= i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1, \quad i^n = i^{n \bmod 4} \\ \overline{a + bi} &= a - bi, \quad |a + bi| = \sqrt{a^2 + b^2}, \quad z\bar{z} = |z|^2 \\ \frac{a + bi}{c + di} &= \frac{(a + bi)(c - di)}{c^2 + d^2} \\ z &= r(\cos \theta + i \sin \theta), \quad z^n = r^n(\cos n\theta + i \sin n\theta) \\ \omega^3 &= 1, \quad 1 + \omega + \omega^2 = 0, \quad \omega^4 = \omega, \quad \omega^5 = \omega^2\end{aligned}$$

Tricks: Reduce powers of i modulo 4; powers of ω modulo 3. Divide complex numbers by multiplying conjugate.

6. Matrices and Determinants

$$\begin{aligned}A : m \times n, \quad B : n \times p &\Rightarrow AB : m \times p \\ \begin{vmatrix} a & b \\ c & d \end{vmatrix} &= ad - bc \\ \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} &= a(ei - fh) - b(di - fg) + c(dh - eg) \\ A^{-1} &= \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0\end{aligned}$$

$$\begin{aligned}\det(AB) &= \det A \det B, \quad \det(A^{-1}) = \frac{1}{\det A}, \quad \det(A^T) = \det A \\ \det(kA) &= k^n \det(A), \quad A \text{ singular} \Rightarrow \det(A) = 0, \quad \text{tr}(A) = \text{diagonal sum}\end{aligned}$$

Tricks: Use $ad - bc$, not $bc - ad$. Proportional rows give determinant zero. For AB , use outer dimensions.

7. Quadratic Equations

$$\begin{aligned}ax^2 + bx + c &= 0, \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad D = b^2 - 4ac \\ D > 0 &: \text{real unequal}, \quad D = 0 : \text{real equal}, \quad D < 0 : \text{complex} \\ \alpha + \beta &= -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}, \quad x^2 - (\alpha + \beta)x + \alpha\beta = 0 \\ \text{reciprocal roots of } ax^2 + bx + c &= 0 : \quad cx^2 + bx + a = 0\end{aligned}$$

Tricks: Equal roots means $D = 0$. Sum has minus sign; product does not. Given one root: substitute.

8. Polynomials and Remainder Theorem

$$\begin{aligned}ax^3 + bx^2 + cx + d &= 0 \\ \alpha + \beta + \gamma &= -\frac{b}{a}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a} \\ \text{remainder by } x - a &= f(a), \quad f(a) = 0 \Rightarrow (x - a) \text{ factor} \\ \text{remainder by } ax + b &= f\left(-\frac{b}{a}\right) \\ x^3 - y^3 &= (x - y)(x^2 + xy + y^2), \quad x^3 + y^3 = (x + y)(x^2 - xy + y^2)\end{aligned}$$

Tricks: For $x + 2$, substitute -2 . For $2x - 3$, substitute $3/2$. Avoid long division.

9. Partial Fractions

$$\begin{aligned}\frac{px + q}{(x - a)(x - b)} &= \frac{A}{x - a} + \frac{B}{x - b} \\ \frac{px + q}{(x - a)^2} &= \frac{A}{x - a} + \frac{B}{(x - a)^2} \\ \frac{px + q}{(x - a)(x^2 + bx + c)} &= \frac{A}{x - a} + \frac{Bx + C}{x^2 + bx + c}\end{aligned}$$

Tricks: Use denominator zeros to isolate coefficients. Repeated factors need each power. Quadratic factor gets numerator $Bx + C$.

10. Sequences and Series

$$\begin{aligned}a_n &= a + (n-1)d, \quad S_n = \frac{n}{2}[2a + (n-1)d], \quad S_n = \frac{n}{2}(a+l) \\a_n &= ar^{n-1}, \quad S_n = \frac{a(r^n - 1)}{r - 1}, \quad S_\infty = \frac{a}{1-r}, \quad |r| < 1 \\AM &= \frac{a+b}{2}, \quad GM = \sqrt{ab}, \quad HM = \frac{2ab}{a+b}, \quad AM \geq GM \geq HM \\ \sum_{k=1}^n k &= \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \\ \sum_{k=1}^n k^3 &= \left[\frac{n(n+1)}{2} \right]^2, \quad 1 + 3 + \cdots + (2n-1) = n^2\end{aligned}$$

Tricks: AP has common difference; GP has common ratio. If a, x, b are in GP, $x^2 = ab$.

11. Permutation and Combination

$$\begin{aligned}n! &= n(n-1) \cdots 1, \quad 0! = 1 \\ {}^nP_r &= \frac{n!}{(n-r)!}, \quad {}^nC_r = \frac{n!}{r!(n-r)!}, \quad {}^nC_r = {}^nC_{n-r} \\ \text{repeated objects: } &\frac{n!}{p!q!r!}, \quad \text{circular arrangements} = (n-1)!\end{aligned}$$

Tricks: Arrangement = permutation. Selection = combination. Together = bundle first. Not together = total - together.

12. Probability

$$\begin{aligned}P(E) &= \frac{n(E)}{n(S)}, \quad 0 \leq P(E) \leq 1, \quad P(E') = 1 - P(E) \\ P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ P(A \cap B) &= 0 \text{ if mutually exclusive, } P(A \cap B) = P(A)P(B) \text{ if independent} \\ P(A|B) &= \frac{P(A \cap B)}{P(B)}, \quad P(X=r) = {}^nC_r p^r q^{n-r}, \quad q = 1 - p\end{aligned}$$

Tricks: At least one = $1 - P(\text{none})$. Dice total 36. Coins total 2^n . Cards total 52. “Or” means union.

13. Binomial Theorem

$$\begin{aligned}(a+b)^n &= \sum_{r=0}^n {}^nC_r a^{n-r} b^r, \quad T_{r+1} = {}^nC_r a^{n-r} b^r \\ n \text{ even : } &T_{n/2+1}, \quad n \text{ odd : } T_{(n+1)/2}, T_{(n+3)/2}\end{aligned}$$

Tricks: Coefficient sum: put $x = 1$. Alternating sum: put $x = -1$. Constant term: exponent of x is zero.

14. Trigonometry Basics

$$\begin{aligned}\sin \theta &= \frac{P}{H}, \quad \cos \theta = \frac{B}{H}, \quad \tan \theta = \frac{P}{B} \\ \cot \theta &= \frac{1}{\tan \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \csc \theta = \frac{1}{\sin \theta} \\ \sin^2 \theta + \cos^2 \theta &= 1, \quad 1 + \tan^2 \theta = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta \\ \sin 30^\circ &= \frac{1}{2}, \quad \sin 45^\circ = \frac{1}{\sqrt{2}}, \quad \sin 60^\circ = \frac{\sqrt{3}}{2} \\ \cos 30^\circ &= \frac{\sqrt{3}}{2}, \quad \cos 45^\circ = \frac{1}{\sqrt{2}}, \quad \cos 60^\circ = \frac{1}{2} \\ \tan 30^\circ &= \frac{1}{\sqrt{3}}, \quad \tan 45^\circ = 1, \quad \tan 60^\circ = \sqrt{3}\end{aligned}$$

Tricks: ASTC. If $\sin \theta = 3/5$, use 3-4-5. If $\tan \theta = 3/4$, then $\sin = 3/5, \cos = 4/5$.

15. Trigonometric Identities

$$\begin{aligned}\sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\ \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\ \tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B} \\ \sin 2A &= 2 \sin A \cos A, \quad \cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1 \\ \tan 2A &= \frac{2 \tan A}{1 - \tan^2 A}, \quad \sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}, \quad \cos^2 \frac{A}{2} = \frac{1 + \cos A}{2} \\ 2 \sin A \cos B &= \sin(A + B) + \sin(A - B) \\ 2 \cos A \cos B &= \cos(A + B) + \cos(A - B) \\ 2 \sin A \sin B &= \cos(A - B) - \cos(A + B) \\ \max(a \sin x + b \cos x) &= \sqrt{a^2 + b^2}\end{aligned}$$

Tricks: $\sin 75 \cos 15 - \cos 75 \sin 15 = \sin 60$. Use $\sqrt{a^2 + b^2}$ for max problems.

16. Trigonometric Equations and Graphs

$$\begin{aligned}\sin x, \cos x : 2\pi, \quad \tan x, \cot x : \pi, \quad \sin(kx), \cos(kx) : \frac{2\pi}{|k|}, \quad \tan(kx) : \frac{\pi}{|k|} \\ \sin x = 0 \Rightarrow x = n\pi, \quad \cos x = 0 \Rightarrow x = \frac{\pi}{2} + n\pi, \quad \tan x = 0 \Rightarrow x = n\pi \\ -1 \leq \sin x \leq 1, \quad -1 \leq \cos x \leq 1, \quad \tan x \in \mathbb{R}\end{aligned}$$

Tricks: Count only in the interval. Period of $\tan(2x)$ is $\pi/2$. If $\sin A = \cos A$ and acute, $A = 45^\circ$.

17. Inverse Trigonometric Functions

$$\begin{aligned}\sin^{-1} x &\in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad \cos^{-1} x \in [0, \pi], \quad \tan^{-1} x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \\ \sin^{-1} x + \cos^{-1} x &= \frac{\pi}{2}, \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2} \\ \tan^{-1} x + \tan^{-1} y &= \tan^{-1} \left(\frac{x + y}{1 - xy} \right) \text{ with quadrant adjustment}\end{aligned}$$

Tricks: Principal branch matters. $\sin^{-1}(\sin x) \neq x$ always.

18. Limits

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1, \quad \lim_{x \rightarrow 0} \frac{\sin ax}{x} = a, \quad \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1 \\ \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} &= \frac{1}{2}, \quad \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x} = 1 \\ \lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a} &= 2a, \quad \lim_{x \rightarrow a} f(x) = f(a) \Rightarrow \text{continuity at } a\end{aligned}$$

Tricks: For $0/0$, factor or rationalize. Substitute only after removing zero denominator.

19. Differentiation

$$\begin{aligned}\frac{d}{dx} c &= 0, \quad \frac{d}{dx} x^n = nx^{n-1}, \quad \frac{d}{dx} \sin x = \cos x, \quad \frac{d}{dx} \cos x = -\sin x \\ \frac{d}{dx} \tan x &= \sec^2 x, \quad \frac{d}{dx} \cot x = -\csc^2 x, \quad \frac{d}{dx} \sec x = \sec x \tan x, \quad \frac{d}{dx} \csc x = -\csc x \cot x \\ \frac{d}{dx} e^x &= e^x, \quad \frac{d}{dx} e^{ax} = ae^{ax}, \quad \frac{d}{dx} a^x = a^x \ln a, \quad \frac{d}{dx} \ln x = \frac{1}{x} \\ (uv)' &= u'v + uv', \quad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \quad \frac{d}{dx} f(g(x)) = f'(g(x))g'(x) \\ m &= \frac{dy}{dx}, \quad y - y_1 = m(x - x_1), \quad m_n = -\frac{1}{m} \\ f'(x) &> 0 : \text{increasing}, \quad f'(x) = 0 : \text{stationary}, \quad f''(x) < 0 : \text{max}, \quad f''(x) > 0 : \text{min}\end{aligned}$$

Tricks: Derivative means slope. Chain rule is the common trap. For $x^2 \sin x$, use product rule.

20. Integration

$$\begin{aligned}\int x^n dx &= \frac{x^{n+1}}{n+1} + C, & \int \frac{1}{x} dx &= \ln|x| + C, & \int e^{ax} dx &= \frac{e^{ax}}{a} + C \\ \int \sin ax dx &= -\frac{\cos ax}{a} + C, & \int \cos ax dx &= \frac{\sin ax}{a} + C \\ \int \sec^2 x dx &= \tan x + C, & \int \csc^2 x dx &= -\cot x + C, & \int \sec x \tan x dx &= \sec x + C \\ \int f(g(x))g'(x) dx &: u = g(x), & \int_a^b f(x) dx &= F(b) - F(a), & A &= \int_a^b y dx\end{aligned}$$

Tricks: Indefinite integral needs $+C$; definite does not. If derivative of inside is present, use substitution.

21. Analytical Geometry: Straight Lines

$$\begin{aligned}d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, & M &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \\ m &= \frac{y_2 - y_1}{x_2 - x_1}, & y &= mx + c, & y - y_1 &= m(x - x_1), & \frac{x}{a} + \frac{y}{b} &= 1 \\ Ax + By + C &= 0, & m_1 &= m_2 \text{ parallel}, & m_1 m_2 &= -1 \text{ perpendicular} \\ d &= \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}\end{aligned}$$

Tricks: Intercepts a, b give $x/a + y/b = 1$. Vertical slope undefined; horizontal slope zero.

22. Circle

$$\begin{aligned}(x - h)^2 + (y - k)^2 &= r^2, & \text{center } (h, k) \\ x^2 + y^2 + 2gx + 2fy + c &= 0, & \text{center } (-g, -f), & r = \sqrt{g^2 + f^2 - c} \\ (x - x_1)(x - x_2) + (y - y_1)(y - y_2) &= 0 & \text{diameter form} \\ xx_1 + yy_1 &= r^2 & \text{tangent to } x^2 + y^2 = r^2\end{aligned}$$

Tricks: Complete square. General form uses $2g, 2f$, not g, f .

23. Conic Sections

$$\begin{aligned}y^2 &= 4ax \Rightarrow \text{focus } (a, 0), & x &= -a \text{ directrix} \\ x^2 &= 4ay \Rightarrow \text{focus } (0, a), & e &= 1 \text{ for parabola} \\ \frac{x^2}{a^2} + \frac{y^2}{b^2} &= 1, & a > b, & \text{major axis} = 2a, & c^2 &= a^2 - b^2, & e &= \frac{c}{a} \\ \frac{x^2}{a^2} - \frac{y^2}{b^2} &= 1, & \text{transverse axis} &= 2a, & c^2 &= a^2 + b^2, & e &= \frac{c}{a} > 1\end{aligned}$$

Tricks: Parabola eccentricity is 1. Ellipse subtracts: $c^2 = a^2 - b^2$. Hyperbola adds: $c^2 = a^2 + b^2$.

24. Linear Inequalities and Linear Programming

$$ax + by \leq c, \quad ax + by = c \text{ boundary line}, \quad Z = ax + by, \quad x \geq 0, y \geq 0 \text{ first quadrant}$$

Tricks: Test $(0, 0)$ to decide side. Maximum/minimum occurs at a vertex. Substitute vertices into Z .

25. Vectors

$$\begin{aligned}\vec{a} &= a_1\hat{i} + a_2\hat{j} + a_3\hat{k}, & |\vec{a}| &= \sqrt{a_1^2 + a_2^2 + a_3^2}, & \hat{a} &= \frac{\vec{a}}{|\vec{a}|} \\ \vec{a} \cdot \vec{b} &= |\vec{a}||\vec{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3 \\ \vec{a} \cdot \vec{b} &= 0 \Rightarrow \text{perpendicular}, & |\vec{a} \times \vec{b}| &= |\vec{a}||\vec{b}|\sin\theta \\ \hat{i} \times \hat{j} &= \hat{k}, & \hat{j} \times \hat{k} &= \hat{i}, & \hat{k} \times \hat{i} &= \hat{j} \\ \cos\theta &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}\end{aligned}$$

Tricks: Dot zero = perpendicular. Cross zero = parallel. Dot gives scalar; cross gives vector.

26. Three-Dimensional Geometry

$$\begin{aligned}d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \\M &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right) \\l &= \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}, \quad l^2 + m^2 + n^2 = 1 \\ \vec{r} &= \vec{a} + \lambda \vec{b}, \quad \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} \\ax + by + cz &= d, \quad \text{normal vector } (a, b, c)\end{aligned}$$

Tricks: Direction cosines always satisfy $l^2 + m^2 + n^2 = 1$. Plane coefficients give normal vector.

27. Variation

$$\begin{aligned}A \propto B &\Rightarrow A = kB, \quad A \propto \frac{1}{B} \Rightarrow A = \frac{k}{B}, \quad AB = k \\A \propto BC &\Rightarrow A = kBC\end{aligned}$$

Tricks: Direct variation: ratio constant. Inverse variation: product constant. Always find k first.

28. Quick MCQ Elimination Rules

- Check sign, domain, quadrant, and positivity before solving.
- Determinant trap: $ad - bc$. Root trap: sum has minus sign.
- Composition order matters. For conics, compare with standard form.
- Probability: distinguish “and” from “or”. Arrangement vs selection matters.
- i^n : reduce modulo 4. ω^n : reduce modulo 3.
- Singular matrix: determinant zero. Equal roots: discriminant zero.
- Remainder by $x - a$: put $x = a$. Coefficient sum: put $x = 1$.
- Constant term: exponent of x equals zero. Perpendicular vectors: dot product zero.